

Time-Series Load Data Analysis for User Power Profiling

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Abstract—In this paper, we present a time-warping power profiling model for smart grid consumers based on real time load data acquired smart meters. It profiles consumers based on their power consumption behaviour using the dynamic time warping (DTW) clustering algorithm. Due to the invariability of signal warping of this algorithm, time-disordered load data can be profiled and consumption features be extracted. Two load types are defined and the related load patterns are extracted for clustering consumption behaviour by DTW. The classification methodology is discussed in detail. To evaluate the performance of the method, we analyze the time-series load data measured by a smart meter in a real case. The results verify the effectiveness of the proposed profiling method with 90.91% true positive rate for load type clustering in the best case.

Index Terms—Power profiling, Privacy awareness, Dynamic time warping (DTW), Smart grid

I. INTRODUCTION

Power consumption is an important variable for scale, performance, and demand management in smart grid [1], [2]. Expansion planning and load forecasting are also based on energy data [3]. Analyzing consumer's power consumption behaviour, known as power profiling, is important for demand forecasting, optimizing resource allocation, and anomaly detecting. The power profile of a consumer shows his/her power consumption patterns over a given period and provides a handy tool for various smart grid applications, e.g., DR and load forecasting [4], [5]. Although more realistic power profiles leads to more accurate demand management, collecting fine-grained metering measurements obtains additional information about consumers' daily life patterns [6]–[8].

Various methods have been used for patterns extraction and classification for power profiling. Most rely on the representative data and statistical averages of the consumption of electrical appliances. In reality, a power profile is not only affected by the load patterns of the electrical appliances, but also by consumers' consumption behavior and contextual factors such as time, location, and environment [9]–[11]. Based on this, a comprehensive power profiling is divided into three profiling concepts, namely user, device, and context [12]. The overall power profile is the resultant of them. Appropriate resources and analysis methods are needed to determine load curve characteristics and identify the consumer's power consumption behaviour.

In this paper, we introduce a model to analyze consumers' consumption behaviours in a realistic environment affected by time factor. In order to build time-invariant consumer level load clustering, we perform load data time series correlation using the dynamic time warping (DTW) algorithm [13]. DTW is selected to measure the similarity between the power usage time series due to its properties, i.e., signal warping invariability and implementation simplicity. By clustering the consumer's load data with the DTW algorithm, load forecasting is performed by correlating the historical load data and time series of the new power usage data. We use the electricity consumption data extracted from the almanac of minutely power dataset version 2 (AMPds2) [14] available at Harvard Dataverse. The AMPds2 is a real-world dataset to capture all three main types of consumption, including electricity, water, and natural gas, collected from a house in the Greater Vancouver metropolitan, BC, Canada, over a long period of time (two years) [14]. Short-term power usage data acquired by real-time smart meters are used to extract load patterns and generate realistic power profiling. The similarities between the short-time electricity consumption data points measured by DTW are used to extract the consumer's load patterns.

The main contributions of this paper are as follows:

- Extracting power consumption patterns by measuring DTW similarity between a consumer's load data time series.
- Power profiling based on the signal warping invariability property of DTW algorithm. Thus time-disordered load data can be used for detecting consumption patterns and load type clustering.

The rest of this paper is organized as follows. In Section II, the preliminaries for our model, e.g., time series clustering and power profiling concepts are described. We introduce our proposed model and explain its stages in Section III. In Section IV, we evaluate the accuracy of our power profiling by analyzing time-disordered load data and discuss its results. Section V has our conclusions and a discussion of future work.

II. PRELIMINARIES AND BACKGROUND

A. Time series classification

1) *Euclidean Distance (ED)*: The ED algorithm is the most common similarity metric. It is appropriate for appli-

cations where there is no direct correlation among distinct features [15]. It uses the straight line distance between two points. Given two time series $P = (p_1, p_2, \dots, p_n)$ and $Q = (q_1, q_2, \dots, q_n)$ in n dimensions, the ED between them is:

$$ED(P, Q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}, \quad (1)$$

where p_i and q_i are the coordinates of P and Q in dimension i . Due to its simplicity, efficiency, and being a distance metric, ED is a popular distance measure for many data mining tasks. Of note, it is only applicable to equal length series with equal dimensions. It is very sensitive to mismatches between the series. For example, in the case of two time series, if there is a slightly delay or time shift in one series the ED between them will be unreasonably large, even though they are exactly the same [16]. This leads to it being inaccurate for classification and clustering applications, despite ease of implementation and time efficiency [17].

2) *Dynamic Time Warping (DTW)*: Compared to ED, DTW is more robust to the similarity computation. DTW is a very popular tool in temporal data mining, and distance comparison is less sensitive to signal transformations such as shifting, uniform amplitude scaling, or uniform time scaling [18]. DTW-based time series similarity measures are less affected by time distortion. It allows elastic transformation of time series to detect similar shapes with different phases. Since DTW is invariant against signal warping, e.g., scaling in the time axis or Doppler effect, it is preferred for pattern matching tasks [19].

To find the distance between two time series $X = (x_1, \dots, x_i, \dots, x_n)$ of length n and $Y = (y_1, \dots, y_j, \dots, y_m)$ of length m , DTW is computed by first finding the best alignment between them. An n -by- m matrix is constructed, in which its (i^{th}, j^{th}) element is equal to $(x_i - y_j)^2$, which represents the cost to align the point x_i of time series X with the point y_j of time series Y . An alignment between the two time series is represented by a warping path, $W = w_1, w_2, \dots, w_K$, in the matrix which has to be contiguous, monotonic, start from the bottom-left corner and end at the top-right corner of the matrix. The warping path has length K , which is

$$\max(n, m) \leq K < (n + m). \quad (2)$$

The k^{th} element of the warping path, W , is $w_k = (i, j)$, where i and j are corresponding indexes from time series X and Y , respectively [20]. The best alignment is given by a warping path through the matrix that minimizes the total cost of aligning its points. The minimum total cost is the DTW distance as follows [16], [21]:

$$DTW(X, Y) = \operatorname{argmin}_{(W)} \sqrt{\sum_{k=1, w_k=(i,j)}^K (x_i - y_j)^2}. \quad (3)$$

To compute the minimum cost alignment, dynamic programming (DP) is used, which increases the complexity of the computation. More critically, DP is a sequential process which makes DTW non-parallelizable. Some techniques have been proposed to parallelize DP [19].

B. Power Profiling

Power profiling includes analytical approaches which lead to predictive maintenance, abnormality, and fault detection. While most consumers show variable power consumption behavior, they typically exhibit a certain amount of repetitiveness [1], [22]. A consumer's consumption behaviour is affected by his/her social, professional, and economic situations, appliances used, time, and environment, e.g., weather and dwelling. These factors can be considered as random or complex phenomenon [23].

Power profiling is also used to predict and balance electricity purchases and sales. Poor estimation could be expensive for a utility provider [24]. Personalizing power management and understanding electrical usage patterns help the utilities to send proper demand signals to the DR systems to balance the electrical load. Normally, consumers are clustered based on their load curves. Consumers in the same cluster have similar consumption patterns. By recognizing the load patterns of a new consumer, he/she is classified into a particular consumer group, in which similar rules are applied. Furthermore, abnormal usage can be detected for this new consumer with little information based on load curves in this class [5]. More over than the load prediction and abnormal usage detection, analyzing the user's power consumption data, used for profiling, may lead to determine the working cycle of each residential electrical appliance [10].

Processing load data makes it possible to determine individual electrical load's patterns (e.g., operating schedules). Therefore, the patterns of normal and abnormal operation are available for utility engineers [22]. These patterns are used in the decision tree-based schemes for fault detection or security management [25]. For instance, an abnormal usage increase in an unusual time (e.g., during midnight or in a vacation time) can be caused by an energy theft attack or an operation fault and should be treated accordingly.

For load management, data processing usually occurs locally in DR systems, including AMI and non-intrusive load monitoring (NILM) systems. A NILM is a fundamental tool for extrapolating in-home activity. It dis-aggregates a consumption data stream into individual load signatures and matches them with reference signatures stored in a database to distinguish operating appliances such as refrigerators, air conditioners, and water heaters. The NILM can be used to identify specific electric device/appliance brands and might even identify malfunctioning appliances [26], [27]. Therefore, power profiling can easily compromise the consumer's security and privacy. Credential extraction by the power analysis attack, authentication compromising by replay attack and masquerading, and privacy breaches are possible security issues with power profiling [28].

III. POWER PROFILING MODEL

In this model, the profiling process is divided into four stages, namely data extraction, load data analysis, load data clustering, and power profile assignment, which are illustrated in Fig. 1. Power consumption is profiled by comparing the time series of the consumer's load data and observing the consumption patterns in the short (one-day) and long-term.

A. Data extraction

To gather power usage data and overview daily usage patterns, we used a real-world dataset of electrical consumption from AMPds2 [14]. The short-term power usage data was extracted by real-time smart meters. The load data was collected by DENT PowerScout 18 units [29], located at the electrical circuit breaker panel, every one minute. To have a time series of data points with a manageable size, we select power usage data (Watt-hour) collected every 10 minutes giving 144 acquisition data points per day. This data is analyzed and the consumption patterns are extracted by comparing different days'. We consider the power data points for one day as a single power usage time series.

B. Load data analysis

The 10-minute power usage data points are time-based random variables and used for pattern extraction. To determine the power consumption patterns, we use DTW algorithm to analyze the consumer's power usage time series. Due to its signal warping invariability and implementation simplicity, DTW is selected for analyzing the power usage time series' similarity in different days. We use *dtw - python* module to implement DTW in Python [30]. Essentially, DTW is a point-to-point (P2P) matching algorithm and enforces temporal consistencies among matched point pairs [31].

C. Load data clustering

A power consumption pattern depends on the consumer's usage behaviour, a device's power consumption characteristics, and contextual factors. Consumers consume differently depending on the time of day and day of week (workday versus weekend/holiday). For instance, the load factor of a refrigerator is influenced by its consumption patterns, which depends on factors such as thermal load, number of door openings, and opening duration [32]. Compare to a workday, the refrigerator shows different consumption patterns on a weekend/holiday, when the number of people at home is different and they behave differently (e.g., different sleeping time and watching TV more). Fig. 2 depicts the refrigerator hourly electricity consumption on a workday and a weekend. The consumption is measured per 10 minutes and there are six data points at each hour. Typically, a refrigerator exhibits a cyclic pattern of power consumption due to duty cycling of its compressor and operates at certain intervals during the day [10].

Since the refrigerator continuously works, we use its power consumption patterns to profile user's power consumption and identify unusual events. In this regard, we use the power

TABLE I: Power load patterns.

Characteristics		Power load patterns	
		Workday load	Weekend/Holiday load
DTW clustering	Euc. distance	1170.67	1244.75
	St. deviation	353.57	402.53
Power usage	Peak (W)	309.24	147
	Mean (W)	48.92	41.36

data points of the refrigerator in three months to train our model and extract the patterns. These patterns are used to classify the consumer's power load. Among these days, the workdays and weekends/holidays are categorized into two separate categories. For each category, the similarity between the power usage time series of each day and the other days' is measured by DTW. The similarity is shown by the DTW distance matrix. A profile is defined for each load type based on its DTW matrix, namely workday load profile (WoLP) and weekend/holiday load profile (WeLP). Based on DTW distance matrix for each profile, a set of patterns is extracted. Each set is an usage time series, in which has minimum Euclidean distance to other days' load data at the same category. In Table I, the characteristics of power load patterns sets, including DTW distance (average and standard deviation) and the power usage (peak and mean) are listed. In WoLP, the extracted load patterns set averagely has 1170.67 DTW distance to other load data time series within 353.57 standard deviation. The load patterns set in WeLP has 1244.75 DTW distance to other load data time series on average in 402.53 standard deviation.

D. Power Profile Assignment

The power profile assignment stage assesses the input power usage data and assigns it to a particular power load type. It measures the similarity between the input load data time series and both load patterns sets by DTW algorithm. Depends on the measured similarity, the load data is categorized as a WoLP or WeLP. For instance, Fig. 3 shows Euclidean distance, warp path, and the straight-line fit between the load data points of the refrigerator measured on a sample day and the load data sequences of the predefined workday. There is 1030 DTW distance between these load data points sets. According to Table I, this sample day is matched to WoLP load type and is profiled as a workday. By recognizing the power consumption patterns and matching them to a load type, we can forecast power load, detect non-technical loss, and apply proper rules to a new consumer.

IV. ANALYSIS AND DISCUSSION

To evaluate our power profiling model, we analyze the time-disordered power usage data and identify the corresponding power load type. Daily acquired power usage data of two sequent months (totally 61 days) is extracted from AMPd2. These days include 44 workdays and 17 weekends (Saturdays and Sundays). For each day, the similarities between its power data sequence and both power load patterns sets are measured by DTW. Depends on the measured distance, a power profile (e.g., WoLP or WeLP) is assigned. Based on Table I, if the

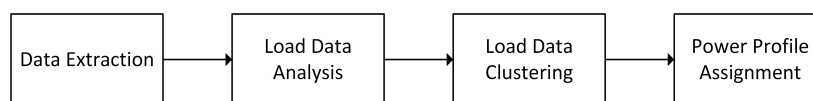
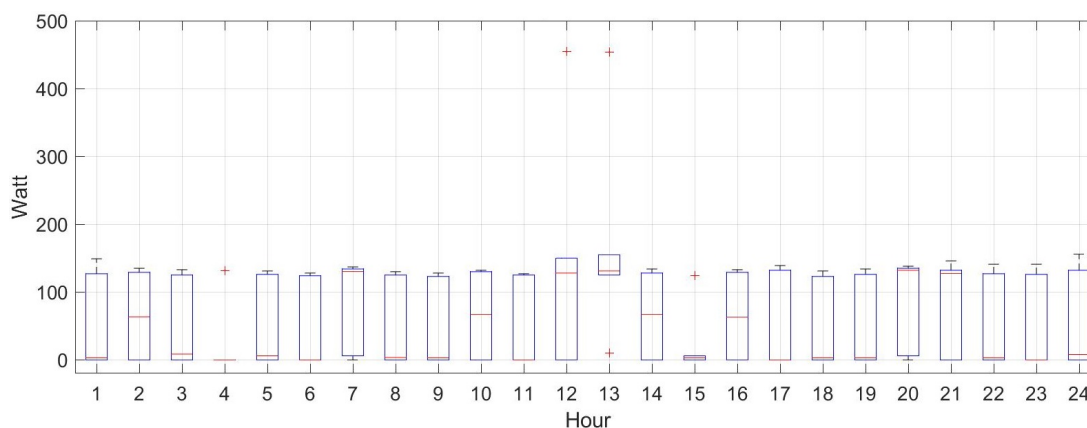
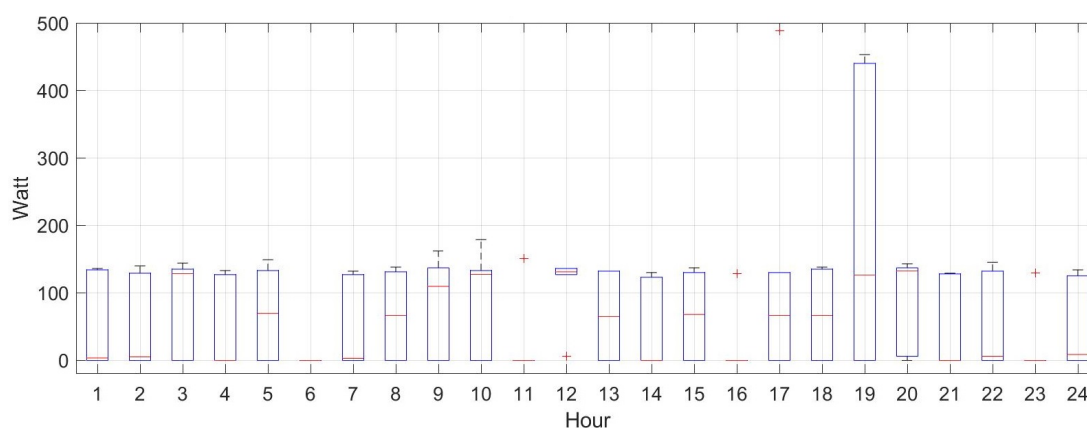


Fig. 1: Power profiling model.



(a) Power usage on a typical workday.



(b) Power usage on a typical weekend.

Fig. 2: Refrigerator hourly electricity consumption, measured per 10 minutes on a typical workday (a) and a typical weekend (b)

measured Euclidean distance to the workday load patterns set is less than 1170.67 the corresponding day is classified as a WoLP, else it is classified as a WeLP. Similarly, it is classified as a WeLP if its power data time series' distance to the weekend patterns set is less than 1244.75. WoLP is assigned if the data sequence has more than 1244.75 distance to the weekend load patterns set.

Table II shows the classification and identification accuracy. The power load clustering is based on the lowest Euclidean distance. Our evaluation shows that using the workday load patterns set leads to better load type clustering. Using this set to measuring Euclidean distance, it correctly identified 40 out of 44 workdays' load type, with the true positive rate (TPR) of 90.91%, and 11 out of 17 weekends' load type, with a true negative rate (TNR) of 64.71%. Although it clusters WoP with an acceptable accuracy, it cannot properly cluster WeP. The

TABLE II: Classification and load type identification accuracy (TPR: true positive rate, FPR: false positive rate).

		Workday load patterns set	Weekend load Patterns set
Classification result	True positive	40	12
	False positive	6	37
	True negative	11	8
	False negative	4	4
	TPR	90.91%	75%
	FPR	35.3%	82.2%
Identification Accuracy	Workday load	90.91%	18.18%
	Weekend load	64.71%	70.58%

false positive rate (FPR) of this clustering (35.3%) confirms this issue. We observed that distance measuring to the weekend load patterns set cannot provide an acceptable accuracy for clustering, especially for the workday load type. This load type

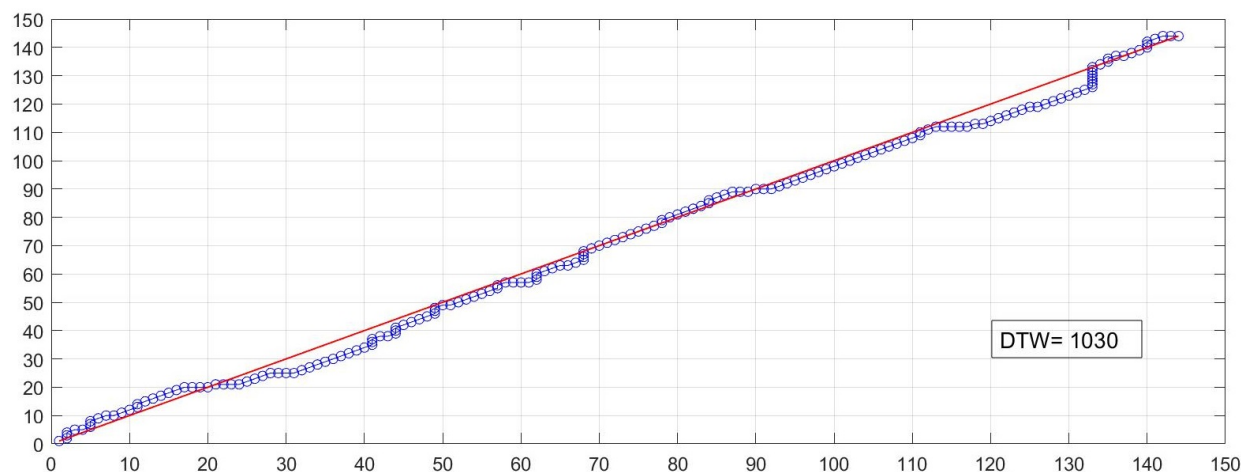


Fig. 3: DTW distance, warp path, and the straight-line fit between the workday load patterns set and a sample day's power consumption data (DTW= 1030).

clustering shows lower TPR, namely 18.18% for workday load type and 75% for weekend load type. Furthermore, it has a high rate of false positive for clustering (FPR= 82.2%).

In observing our results, it is important to note that each load patterns set shows better clustering for its related load type. For instance, a WoLP is identified in 90.91% by DTW similarity to the workday load patterns set and in 18.18% by DTW similarity to the weekend load patterns set. Measuring DTW similarity, a WeLP is identified in 70.58% by the weekend load patterns set and in 64.71% by the workday load patterns set. On the basis of this analysis, extra factors are needed for more accuracy in profiling. In this regard, daily load factor of the user is a proper choice. We plan to add it to load profiles to enhance power profiling and decrease the FPR in clustering based on DTW similarity.

V. CONCLUSION

In this work, we introduced a time-disordered tolerated power profiling model for smart grid consumers. It uses DTW algorithm to profile a consumer's power consumption behaviour and analyzes the load data provided by smart meters in a DR system. DTW clustering algorithm is used for extracting power consumption features and cluster load types. To have real-world data, we used electricity consumption data extracted from AMPd2 dataset. Two load type profiles are defined for workdays and weekends, namely WoLP and WeLP. For each profile, a load patterns set was extracted and the consumers' load types are clustered by measuring their DTW similarity to the introduced patterns sets. The evaluation showed that DTW clustering based on workday load patterns has a better result. This solution can be extended for power profiling several consumers based on their different appliances' power consumption data time series. For future, we plan to add daily load factor to each load profile to enhance the clustering accuracy and decrease FPR.

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REFERENCES

- [1] X. Ma, Z. Du, and J. Liu. Program power profiling based on phase behaviors. *Sustainable Computing: Informatics and Systems*, 19:341–350, 2018.
- [2] D. Toffanin. Generation of customer load profiles based on smart-metering time series, building-level data and aggregated measurements. 2016.
- [3] M. Meliani, A. E. Barkany, I. E. Abbassi, A. M. Darcherif, and M. Mahmoudi. Energy management in the smart grid: State-of-the-art and future trends. *International Journal of Engineering Business Management*, 13:18479790211032920, 2021.
- [4] M.F. Elahe, M. Jin, and P. Zeng. Review of load data analytics using deep learning in smart grids: Open load datasets, methodologies, and application challenges. *International Journal of Energy Research*, 45(10):14274–14305, 2021.
- [5] Y. Wang, Q. Chen, C. Kang, M. Zhang, K. Wang, and Y. Zhao. Load profiling and its application to demand response: A review. *Tsinghua Science and Technology*, 20(2):117–129, 2015.
- [6] Y. Gong, Y. Cai, Y. Guo, and Y. Fang. A privacy-preserving scheme for incentive-based demand response in the smart grid. *IEEE Transactions on Smart Grid*, 7(3):1304–1313, 2015.
- [7] M. D. Firoozjaei, A. H. Lashkari, and A. A. Ghorbani. Memory forensics tools: a comparative analysis. *Cyber Security Technology*, 2022.
- [8] S. Ghosh, U. Chatterjee, D. Chatterjee, R. Masburah, D. Mukhopadhyay, and S. Dey. Demand manipulation attack resilient privacy aware smart grid using pufs and blockchain. In *International Conference on Applied Cryptography and Network Security*, pages 252–275. Springer, 2021.
- [9] L. Chuan and A. Ukil. Modeling and validation of electrical load profiling in residential buildings in singapore. *IEEE Transactions on Power Systems*, 30(5):2800–2809, 2014.
- [10] F. Issi and O. Kaplan. The determination of load profiles and power consumptions of home appliances. *Energies*, 11(3):607, 2018.
- [11] M. D. Firoozjaei, R. Lu, and A. A. Ghorbani. An evaluation framework for privacy-preserving solutions applicable for blockchain-based internet-of-things platforms. *Security and Privacy*, 3(6):e131, 2020.
- [12] T. Kisielewicz, S. Stanek, and M. Zytnewski. A Multi-Agent Adaptive Architecture for Smart-Grid-Intrusion Detection and Prevention. *Energies*, 15(13):4726, 2022.
- [13] M. Muller. *Dynamic Time Warping*, pages 69–84. Springer Berlin Heidelberg, 2007.

- [14] S. Makonin, B. Ellert, I. V. Bajić, and F. Popowich. Electricity, water, and natural gas consumption of a residential house in Canada from 2012 to 2014. *Scientific data*, 3(1):1–12, 2016.
- [15] F. Iglesias and W. Kastner. Analysis of Similarity Measures in Times Series Clustering for the Discovery of Building Energy Patterns. *Energies*, 6(2):579–597, 2013.
- [16] R. J. Kate. Using Dynamic Time Warping Distances as Features for Improved Time Series Classification. *Data Mining and Knowledge Discovery*, 30(2):283–312, 2016.
- [17] C. A. Ratanamahatana and E. Keogh. Making Time-series Classification More Accurate Using Learned Constraints. In *Proceedings of the 2004 SIAM international conference on data mining*, pages 11–22. SIAM, 2004.
- [18] C. Cassisi, P. Montalto, M. Aliotta, A. Cannata, and A. Pulvirenti. Similarity Measures and Dimensionality Reduction Techniques for Time Series Data Mining. *Advances in data mining knowledge discovery and applications (InTech, Rijeka, Croatia, 2012)*, pages 71–96, 2012.
- [19] X. Cai, T. Xu, J. Yi, J. Huang, and S. Rajasekaran. DTWNet: a dynamic time warping network. *Advances in neural information processing systems*, 32, 2019.
- [20] S. Salvador and P. Chan. FastDTW: Toward accurate dynamic time warping in linear time and space. *Intelligent Data Analysis*, 11(5):561–580, 2007.
- [21] Z. Zhang, R. Tavenard, A. Bailly, X. Tang, P. Tang, and T. Corpetti. Dynamic time warping under limited warping path length. *Information Sciences*, 393:91–107, 2017.
- [22] Y. Zhang, T. Huang, and E. F. Bompard. Big data analytics in smart grids: a review. *Energy informatics*, 1(1):1–24, 2018.
- [23] Q. Zhao, H. Li, X. Wang, T. Pu, and J. Wang. Analysis of users' electricity consumption behavior based on ensemble clustering. *Global Energy Interconnection*, 2(6):479–488, 2019.
- [24] A. Sauhats, R. Varfolomejeva, O. Lmkevis, R. Petrecenko, M. Kunickis, and M. Balodis. Analysis and prediction of electricity consumption using smart meter data. In *2015 IEEE 5th International Conference on Power Engineering, Energy and Electrical Drives (POWERENG)*, pages 17–22. IEEE, 2015.
- [25] M. D. Firoozjaei, J. Park, and H. Kim. Detecting False Emergency Requests Using Callers' Reporting Behaviors and Locations. In *2016 30th International Conference on Advanced Information Networking and Applications Workshops (WAINA)*, pages 243–247. IEEE, 2016.
- [26] M. D. Firoozjaei, N. Mahmoudiyar, Y. Baseri, and A. A. Ghorbani. An evaluation framework for industrial control system cyber incidents. *International Journal of Critical Infrastructure Protection*, page 100487, 2022.
- [27] M. A. Lisovich, D. K. Mulligan, and S. B. Wicker. Inferring personal information from demand-response systems. *IEEE Security & Privacy*, 8(1):11–20, 2010.
- [28] M. D. Firoozjaei, J. Yu, and H. Kim. Privacy Preserving Nearest Neighbor Search Based on Topologies in Cellular Networks. In *2015 IEEE 29th International Conference on Advanced Information Networking and Applications Workshops*, pages 146–149. IEEE, 2015.
- [29] DENT Instruments. PowerScout Series, NETWORKED POWER METERS. <https://www.pc-s.com/pdf/dent-powerscout-powermeters-submeters-series.pdf>, 2010.
- [30] T. Giorgino. Computing and visualizing dynamic time warping alignments in R: the dtw package. *Journal of statistical Software*, 31:1–24, 2009.
- [31] J. Zhao and L. Itti. shapedtw: Shape dynamic time warping. *Pattern Recognition*, 74:171–184, 2018.
- [32] J. M. Belman-Flores, D. Pardo-Cely, M. A. Gómez-Martínez, I. Hernández-Pérez, D. A. Rodríguez-Valderrama, and Y. Heredia-Aricapa. Thermal and energy evaluation of a domestic refrigerator under the influence of the thermal load. *Energies*, 12(3):400, 2019.



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